

Customer Segmentation

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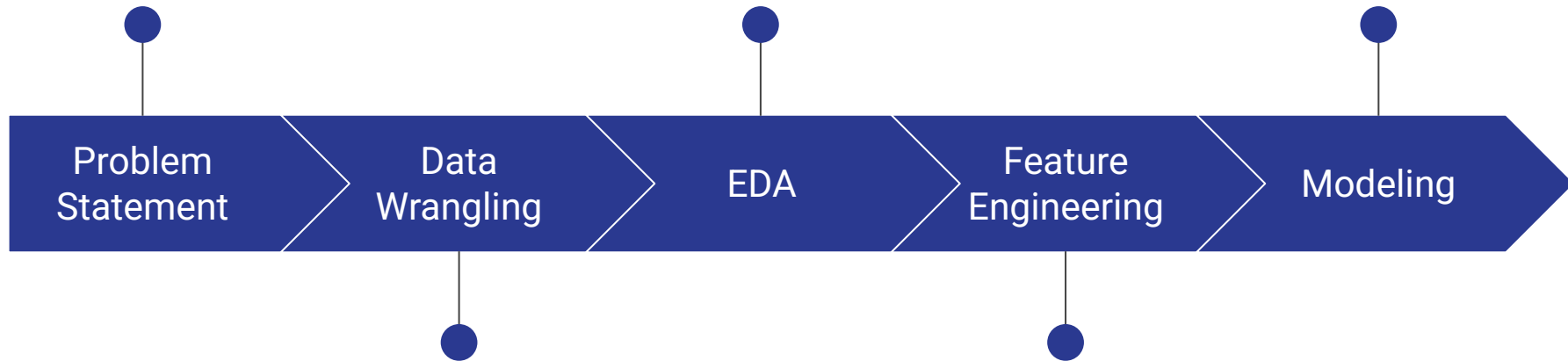
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1) DSM Method

Identifying Correct
Problem to Solve

Exploratory Data
Analysis and testing
Hypothesis

Finding best machine
learning model with
optimal values



Cleaning the data by
filling Null values and
changing data types

Add new attributes
and make data ready
for modeling

2) Data Source

Data Source

For this project, I have used a dataset from [Kaggle.com](https://www.kaggle.com) website.

Here is a link to the dataset on [Kaggle.com](https://www.kaggle.com) :

<https://www.kaggle.com/code/farzadnekouei/customer-segmentation-recommendation-system>

3) Problem Identification

DSM Step 1: Problem Identification

To expand business, an online retailing company wants to have a comprehensive analysis of the customers and products. Therefore, the company wants to segment customers based on shopping behavior and then recommend them most related products.

The online retail company's marketing team, executive sales manager, and the IT Development team are stakeholders for this project.

4) Data Wrangling

DSM Step 2: Data Wrangling

The received dataset has so many Null values. Most of these Null values are 'Unit Price' of an item which is recorder as 0. As the first step of Data Wrangling, I will be removing or imputing values for those rows with Null values.

Data Type is another problem of this dataset. 'Date' column is of string type, which needs to be converted to datetime type.

There are some outliers as well, but most of them come from wrong inputs. Simply, I will be removing the outliers.

5) EDA - Exploratory Data Analysis

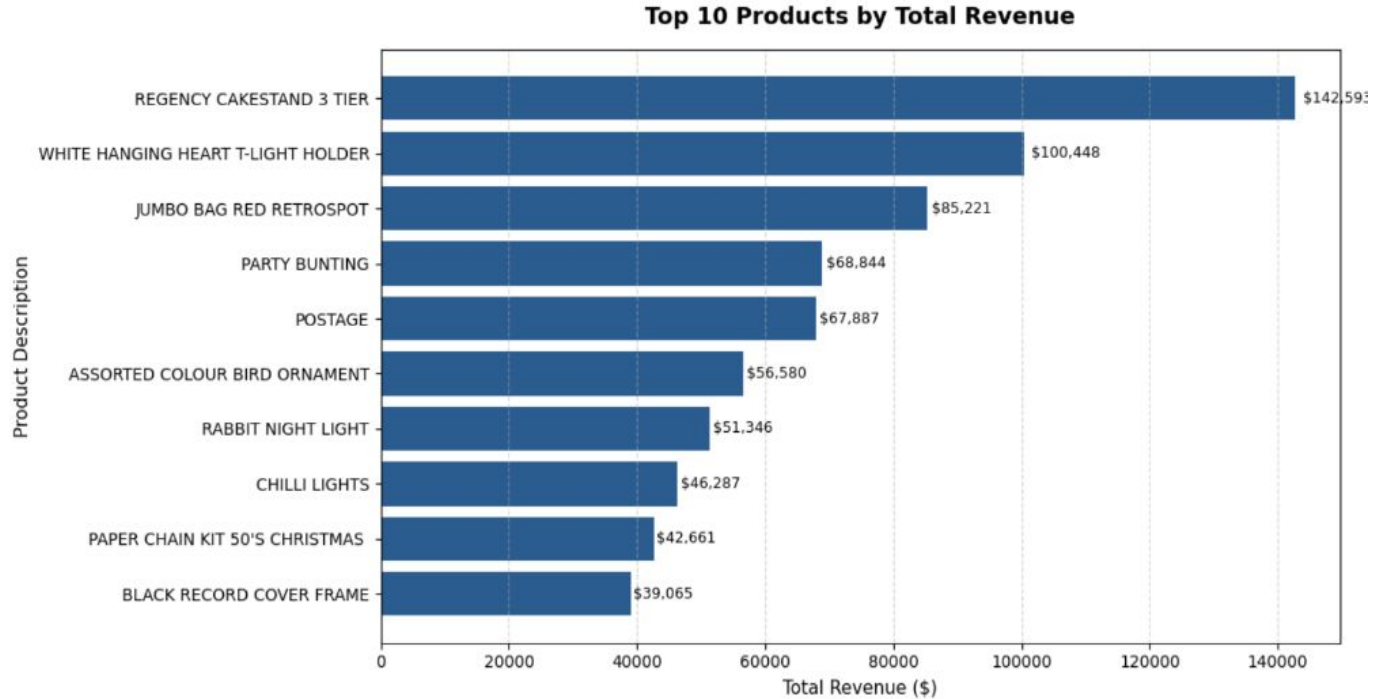
DSM Step 3: EDA - Exploratory Data Analysis

I start EDA by drawing bar charts, scatterplots, heatmaps, and line plots.

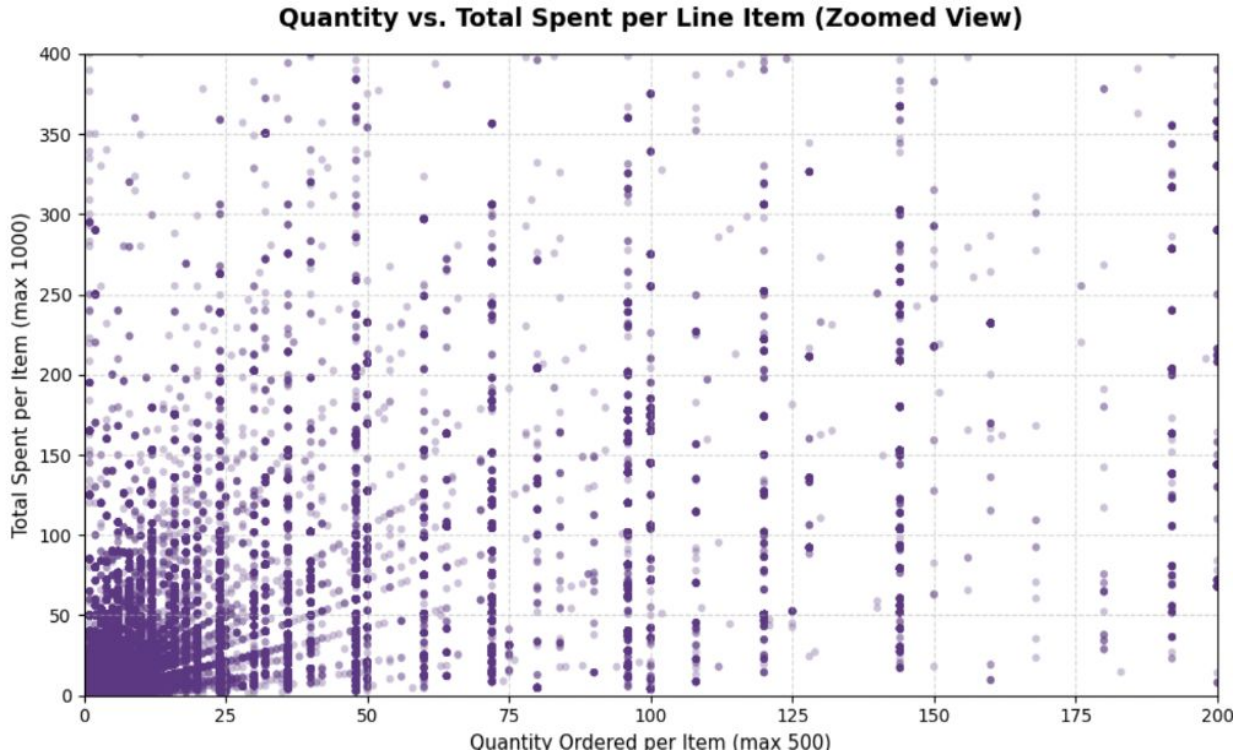
Then a Null Hypothesis and Alternative Hypothesis will be tested.

Discovering correlation between attributes will be considered as well.

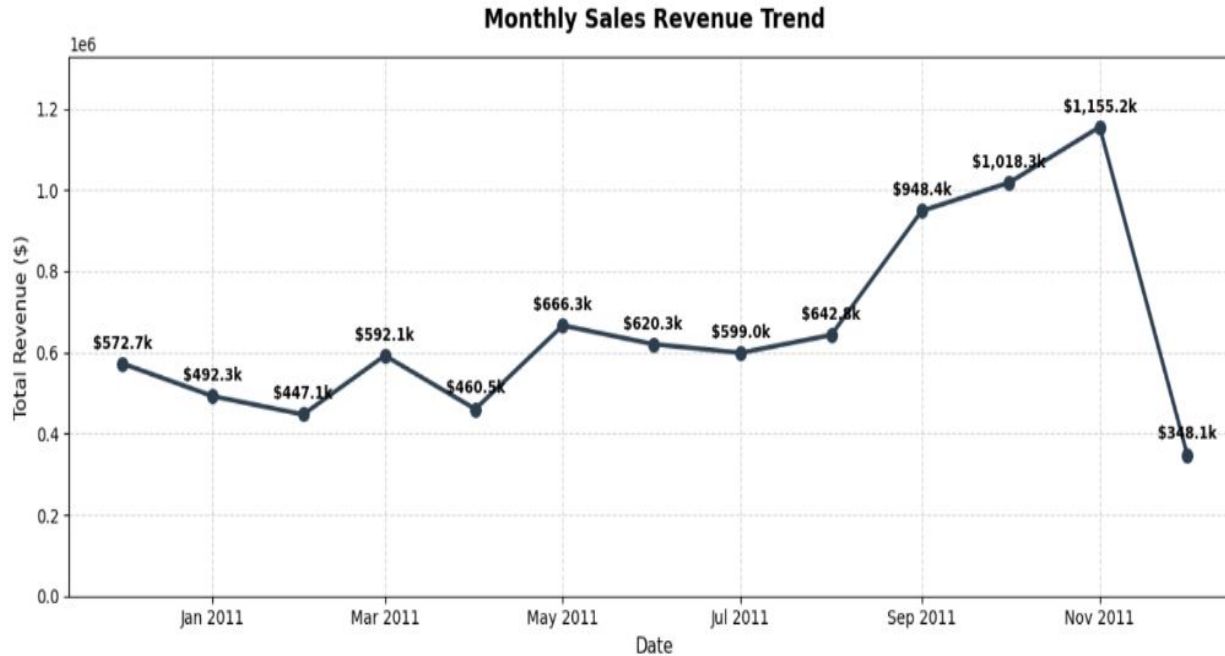
A Bar Chart showing top 10 best selling products



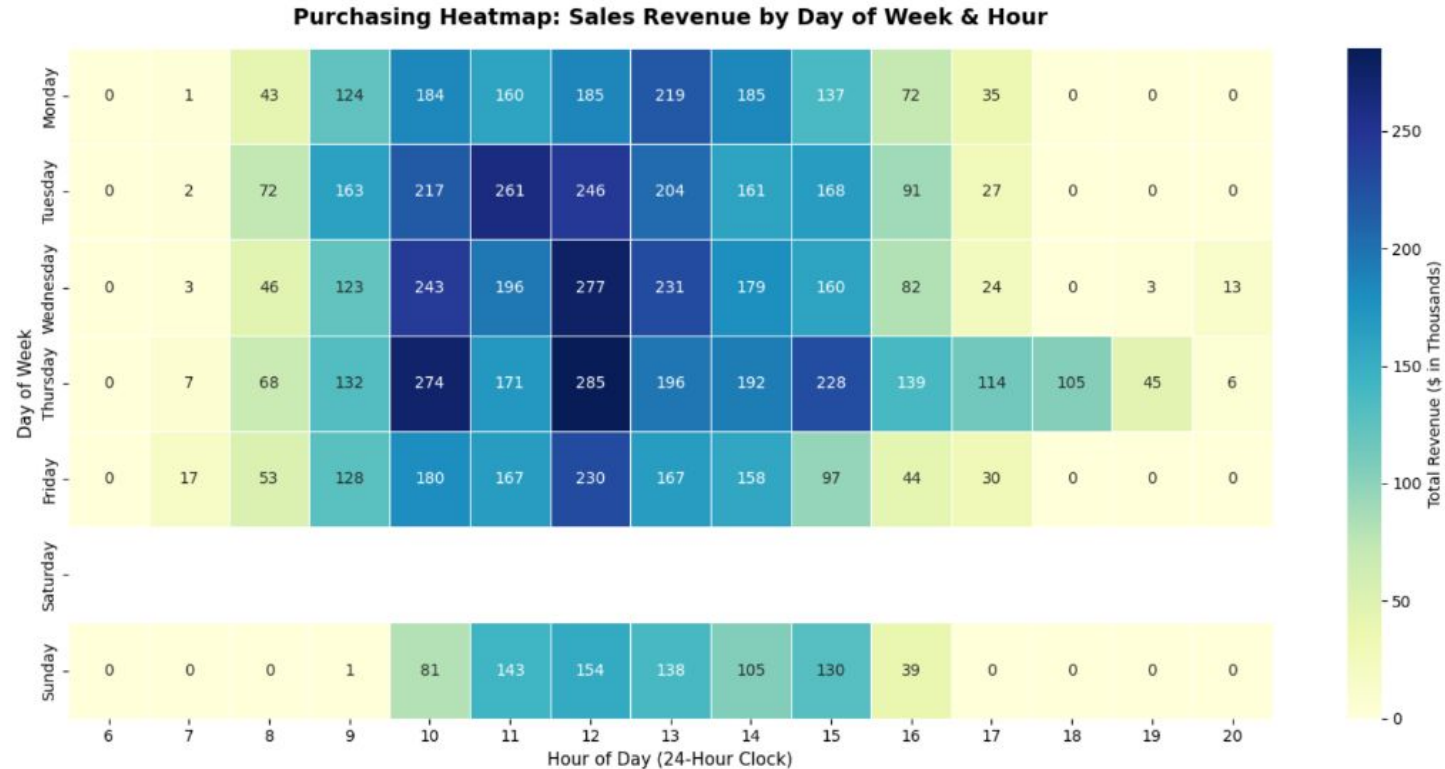
A Scatterplot shows Quantity vs Total Spent per Line Item



A Line-Plot shows monthly sales revenue trend for the online store



A Heatmap which shows Sales Revenue by Day of Week & Hour



6) Feature Engineering

DSM Step 4: Feature Engineering

In this step, we preprocess our dataframe and try to add new attributes, and remove those attributes which are unnecessary for a machine learning model to learn. For example, I removed 'Country' column since all transactions happened in United Kingdom.

I have added three columns to my dataframe. Those columns are Recency, Frequency, and Monetary which are very important for this type of analysis.

7) Machine Learning Modeling

DSM Step 5: Machine Learning Modeling

Customer Segmentation is an Unsupervised Machine Learning problem.

I have worked on 2 different unsupervised machine learning models:

- 1) K-Means Clustering
- 2) Gaussian Mixture Model

K-Means Clustering:

I have found Silhouette score as 7 which means there are 7 clusters for this online retail store customers.

An Excel sheet is exported and provided at my GitHub repo with all customers assigned to associated clusters.

GMM - Gaussian Mixture Model:

This model has soft clustering. In other words, it assigns a set of probabilities to each customer. For example, a customer can belong to Electronics segment by 70% and Household segment by 30%.

8) Data Storytelling

DSM Step 6: Data Storytelling

We have two results for this project. One is from K-Means Clustering and the other is from GMM.

An Excel sheet is exported and saved at my GitHub repo which shows each customer and associated cluster.

Also, as the result of GMM, a set of probabilities is provided for each customer which can be found in my GitHub repo as well.